EXPLAINABLE - DRIFT-AWARE - COST-ACCOUNTED

TabML-Guard Report — Bike Sharing Demand

data/bike_sharing.csv · 17,379 rows × 12 features · regression

RECOMMENDED MODEL

lightgbm_random

RMSE (HELD-OUT)

93.6287

AT A GLANCE

- Tuning: real gain
- Drift detected

Headline finding

lightgbm_random is the recommended model, scoring 93.6287 rmse on a strictly held-out, time-ordered test set.

Recommendation

Tuning improved the score by 1.9% over defaults (best strategy: random). At an equal budget Bayesian search did not beat random search - on a space this size, random is the pragmatic default. Both the inputs and the accuracy moved over time: schedule retraining.

This report was produced automatically by the TabML-Guard pipeline. Every number below comes from a single run on a strictly held-out, time-ordered test split - no test data was used for model selection or tuning.

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STAGE 0

Dataset, profile and configuration

The pipeline begins by profiling the raw table and deciding, from the target column alone, whether this is a classification or a regression problem. When a time column is present the split is chronological - the model trains on the past and is scored on the future.

Rows	17,379
Features	12
Target column	cnt
Detected task	regression
Missing values	0
Time column	dteday
Time span	2011-01-01 -> 2012-12-31
Target statistics	min=1.0000 max=977.0000 mean=189.4631 std=181.3876

Configuration used

This is the entire dataset-specific input to the run. Everything else is generic pipeline code.

```
dataset.source        csv
dataset.name          -
dataset.path          data/bike_sharing.csv
dataset.target        cnt
dataset.time_col      dteday
dataset.drop_cols     ['instant', 'casual', 'registered']
split                 train=0.6  test=0.4
models.boosters       ['xgboost', 'lightgbm', 'catboost', 'random_forest']
models.stacking       True
tuning.budget_trials  40
tuning.cv_splits      3
drift.n_chunks        10
drift.retrain         sliding
random_state          42
```

STAGE 1

Regularized linear baselines

Before reaching for an ensemble, the pipeline establishes what a well-regularized linear model can do. This is the honesty check: if a linear model is competitive, the extra complexity of boosting has to justify itself.

model	rmse	mae	r2	fit
linear	**186.0113**	**141.0464**	**0.2729**	**0.0s**
lasso	186.0123	141.0461	0.2729	1.0s
ridge	186.0139	141.0465	0.2729	0.0s
elasticnet	186.0257	141.0467	0.2728	1.9s

the 4 regularizers span only 0.0145 rmse (best 'linear' 186.0113) — with abundant rows and few features, penalties barely move the fit at the default strength; Lasso path: at alpha=10 only 6/12 features survive (sparsity)

STAGE 2

Ensemble and boosting models

Gradient-boosted trees plus a stacking meta-model, all scored through the same evaluation path as the baselines so the leaderboard rows are directly comparable.

model	rmse	mae	r2	fit
lightgbm	**94.6518**	**66.4706**	**0.8117**	**0.4s**
xgboost	95.7901	67.5324	0.8072	0.3s
stacking	96.2331	67.6730	0.8054	8.4s
random_forest	107.3651	76.2367	0.7578	0.8s
catboost	109.9642	78.7150	0.7459	0.7s
linear	186.0113	141.0464	0.2729	0.0s
lasso	186.0123	141.0461	0.2729	1.0s
ridge	186.0139	141.0465	0.2729	0.0s
elasticnet	186.0257	141.0467	0.2728	1.9s

Best ensemble: **lightgbm**. The strongest single booster (**lightgbm**) is the one carried into the tuning comparison in Stage 3.

STAGE 3

Hyperparameter optimization: benefit vs cost

Four strategies compete on **lightgbm**: library defaults, an exhaustive coarse grid, random search, and Bayesian optimization (Optuna TPE). Random and Bayesian are given an **equal budget** of 40 trials, which makes them a fair head-to-head. Crucially, the table reports not just the score but what each strategy *cost*.

strategy	test rmse	wall clock	model fits	cv score
default	95.4236	0.2s	1	-
grid	98.7340	3.1s	54	-70.1296
random	**93.6287**	**3.3s**	**120**	**-69.5964**
bayes	95.3126	7.6s	120	-69.0437

The *cv score* column is the internal cross-validated score each search optimized (scikit-learn's convention is higher-is-better, so error metrics appear negated). The *test* column is the winning configuration refit once and scored on the untouched held-out period - that is the number to trust.

> **Is tuning worth it?**
>
> best tuner 'random' reached rmse=93.6287 vs default 95.4236 (+1.7949); at equal budget, random search won (rmse: bayes 95.3126 @ 7.62s vs random 93.6287 @ 3.3s)

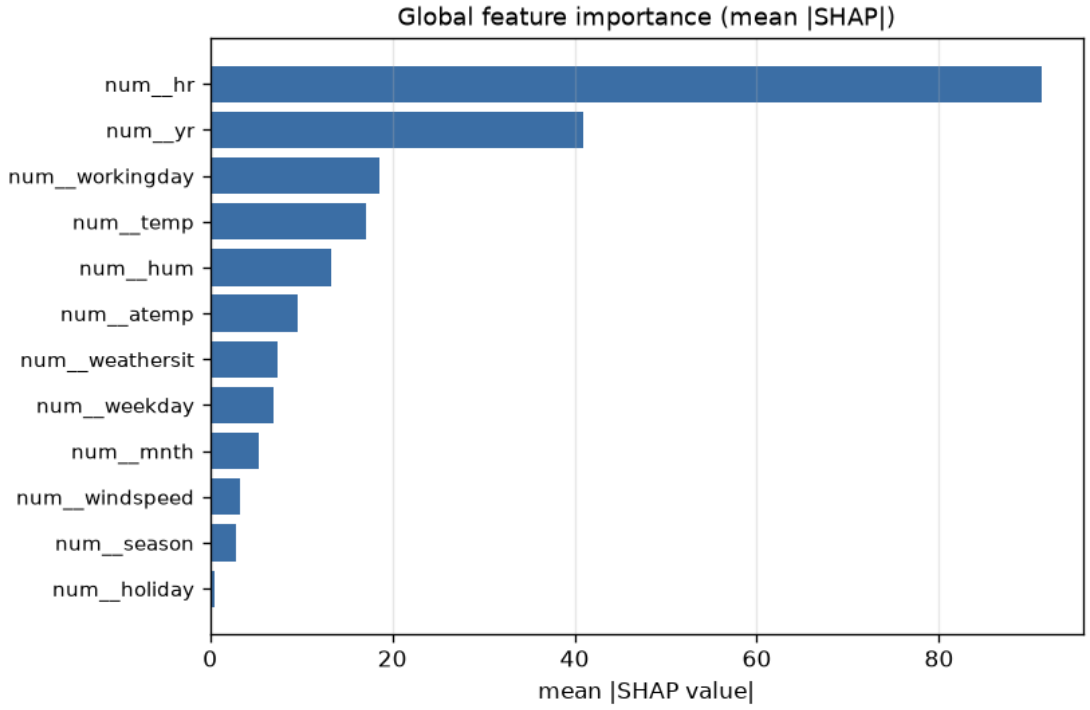
STAGE 4

Explainability and prediction reliability

SHAP attributions for **lightgbm_random**, using the explainer matched to the model family (tree, linear, or model-agnostic fallback). Global importance answers *what drives this model overall*; the local example shows how a

single prediction was reached.

feature	mean |SHAP|
num__hr	91.21615
num__yr	40.89955
num__workingday	18.55817
num__temp	17.18202
num__hum	13.33537
num__atemp	9.67769
num__weathersit	7.38602
num__weekday	7.01741
num__mnth	5.32998
num__windspeed	3.23738
num__season	2.81621
num__holiday	0.54273



Global feature importance (mean |SHAP|)

reliability analysis unavailable.

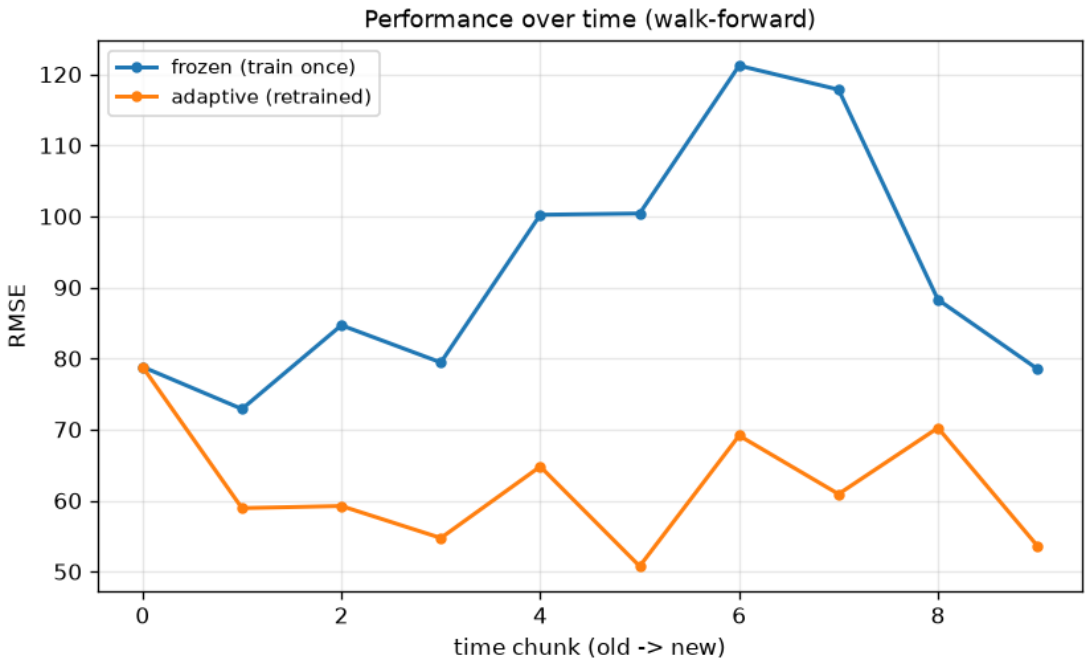
STAGE 5

Concept drift and statistical significance

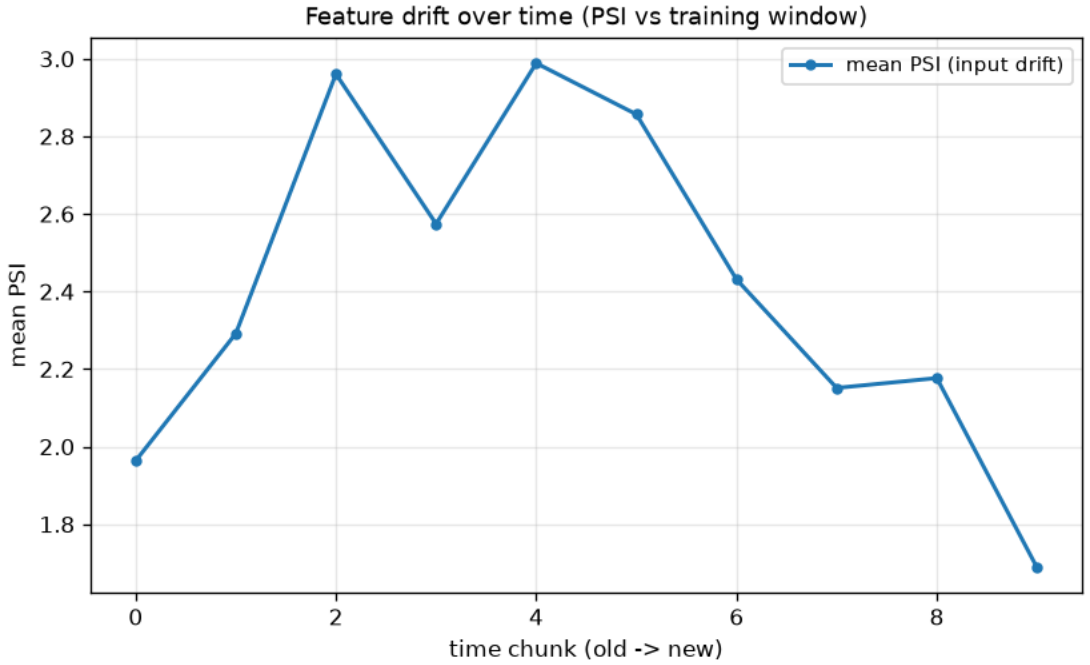
The held-out period is replayed in 10 chronological chunks. A **frozen** model (trained once, never updated) is compared against an **adaptive** model that predicts each chunk *before* learning from it - so there is no look-ahead. Retraining mode: **sliding**.

chunk	rows	frozen	adaptive	mean PSI
0	696	78.8167	78.8167	1.964
1	696	72.8875	58.9285	2.292
2	695	84.6730	59.2311	2.961
3	695	79.4362	54.7265	2.574
4	695	100.2272	64.7898	2.989
5	695	100.4088	50.7559	2.857
6	695	121.2154	69.1438	2.431
7	695	117.8425	60.9149	2.151
8	695	88.2789	70.2313	2.177
9	695	78.5306	53.6189	1.689

Chunk 1 is a built-in consistency check: both models are identical there by construction, since neither has seen the chunk yet.



Frozen vs adaptive performance across the held-out period



Input drift (PSI) against the training window

Mean score, frozen model	`92.2317`
Mean score, adaptive model	`62.1157`
First third -> last third	`-  ->  -`
Performance worsened	`no`
Worst-drifting feature	`-`
Peak mean PSI	`2.989`

Drift verdict

Frozen-model RMSE declined across time (78.817 -> 78.531; drift magnitude +16.092). Peak input drift (mean PSI) = 2.989 (notable). A sliding-retrained model beat the frozen one (RMSE 62.116 vs 92.232, gap 30.116 and the difference is statistically significant). Champion vs linear baseline: significant (p=0.0).

Statistical significance

A score difference is only meaningful if it survives a significance test. Paired predictions on the same held-out rows are compared with the Wilcoxon signed-rank test. A p-value below 0.05 means the difference is unlikely to be chance; above it, the two models are statistically indistinguishable - which is itself worth reporting.

comparison	better model	p-value	conclusion
champion vs runner up	lightgbm_random	`0.702`	`not significant`
champion vs linear baseline	lightgbm_random	`0`	`significant`
static vs adaptive	adaptive	`0`	`significant`

champion vs runner up: lightgbm_random has lower absolute error; but not significantly (p=0.702).

champion vs linear baseline: lightgbm_random has lower absolute error; the difference is statistically significant (p=0).

static vs adaptive: adaptive has lower absolute error; the difference is statistically significant (p=0).

APPENDIX A

Scope and limitations

Tuning is measured, not assumed

The point of Stage 3 is to quantify the cost/benefit of tuning, not to prove that tuning always wins. On well-conditioned datasets a regularized baseline can match a tuned booster, and this report says so when it happens.

Drift is detected, not explained

PSI and prequential error are cheap, reliable indicators that a distribution has moved. Neither attributes a cause; deeper causal drift analysis is out of scope.

Single-machine scale

The pipeline targets datasets that fit in memory on one machine. It is not a distributed training system.

One split, one seed

Results come from a single temporal split at a fixed random seed. That is the right choice for a time-ordered problem - shuffled k-fold would leak the future - but it means confidence intervals across resamples are not reported, which is why the significance tests above matter.

APPENDIX B

Glossary

ROC-AUC	Probability a random positive is ranked above a random negative. 0.5 is chance, 1.0 is perfect. Insensitive to class imbalance.
RMSE / MAE	Average prediction error in the target's own units. Lower is better; RMSE punishes large misses harder than MAE.
R-squared	Share of the target's variance the model explains. 0 means no better than predicting the mean.
Regularization (L1/L2)	A penalty on coefficient size that trades a little bias for less variance. L1 can zero coefficients outright, performing feature selection; L2 shrinks them smoothly.
Stacking	Training a small meta-model on the base models' predictions so it can learn which to trust where.
Bayesian optimization	Tuning that builds a surrogate model of the search space to choose each next trial, rather than sampling blindly.
SHAP value	A feature's signed contribution to one prediction, derived from cooperative game theory. Contributions sum to the prediction.
PSI	Population Stability Index: how far a feature's distribution has moved from its training reference. Rules of thumb: below 0.1 stable, 0.1-0.25 moderate, above 0.25 material.
Prequential evaluation	Test-then-train: each chunk is predicted before the model is allowed to learn from it, so no future information leaks.
McNemar's test	A paired test for two classifiers on identical rows, based only on the cases where they disagree.

Produced by the TabML-Guard pipeline.

~ Ayush Gole